Template Week 6 – Networking

Student number: 591491

Assignment 6.1: Working from home

Screenshot installation openssh-server:

```
tim@DellG15Tim:~$ sudo apt list --installed | grep ssh-server

WARNING: apt does not have a stable CLI interface. Use with caution in scrip
ts.

openssh-server/noble-updates,now 1:9.6p1-3ubuntu13.14 amd64 [installed]
tim@DellG15Tim:~$
```

Screenshot successful SSH command execution:

```
PS C:\Users\timni> ssh tim@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is SHA256:iVuwR3F5ZLBWa5zPfgZmBQuMNclNDi+tD18grxV0az
E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
tim@localhost's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x8
6_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8
s
   just raised the bar for easy, resilient and secure K8s cluster deployment
.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
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This message is shown once a day. To disable it please create the
/home/tim/.hushlogin file.
tim@DellG15Tim:~$
```

Screenshot successful execution SCP command:

```
C:\Users\timni\Downloads>scp .\yes.mov tim@localhost:/home/tim/
tim@localhost's password:
yes.mov                                                   100% 4852KB  41.2MB/s   00:00

C:\Users\timni\Downloads>
```

Screenshot remmina:

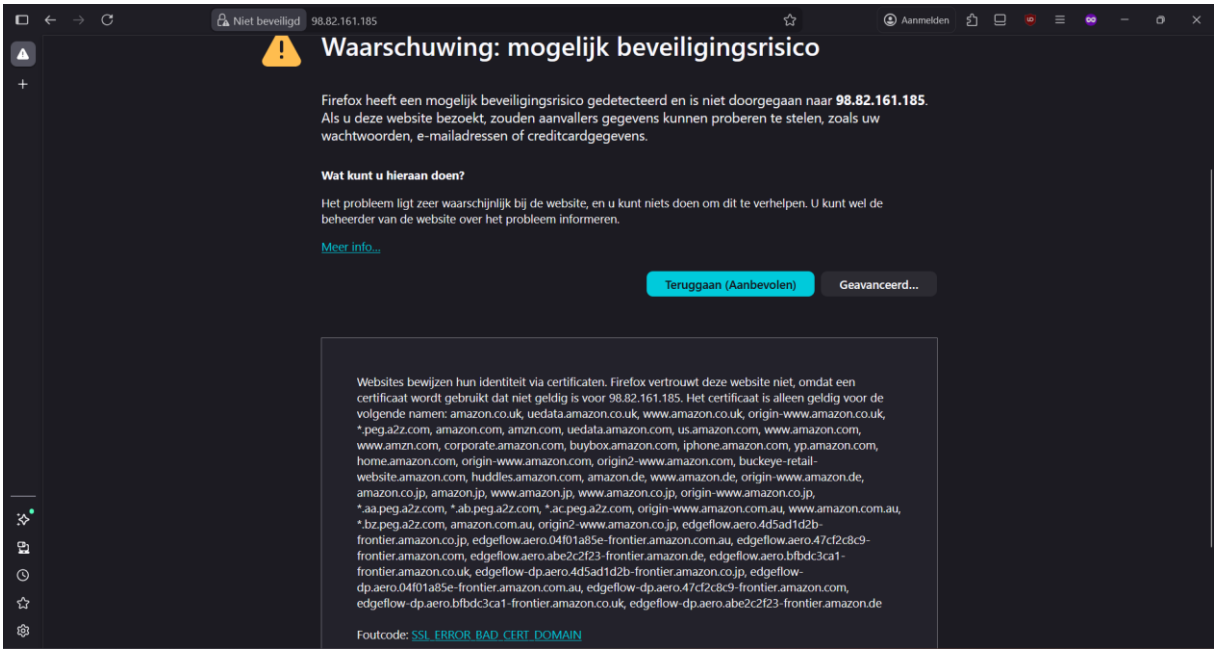
```
root@DellG15Tim: /mnt/c
root@DellG15Tim:/mnt/c# dir
$Recycle.Bin              Program\ Files                 Windows
$SysReset                 Program\ Files\ (x86)          XboxGames
Apps                      ProgramData                    backup
Config.Msi                PropertyVISUALSTUDIO2017FOLDER e-logo
Dell                      Recovery                       hiberfil.sys
Documents\ and\ Settings  SAXION                         inetpub
Drivers                   System\ Repair                 langpacks
DumpStack.log.tmp         System\ Volume\ Information    pagefile.sys
OneDriveTemp              Temp                           swapfile.sys
PerfLogs                  Users
root@DellG15Tim:/mnt/c#
```

Assignment 6.2: IP addresses websites

Relevant screenshots nslookup command:

```
Command Prompt

C:\Users\timni>nslookup amazon.com
Server:  router.domain_not_set.invalid
Address:  192.168.1.254

Non-authoritative answer:
Name:    amazon.com
Addresses:  98.82.161.185
          98.87.170.74
          98.87.170.71


C:\Users\timni>nslookup google.com
Server:  router.domain_not_set.invalid
Address:  192.168.1.254

Non-authoritative answer:
Name:    google.com
Addresses:  2a00:1450:400e:803::200e
          142.250.179.206


C:\Users\timni>nslookup one.one.one.one
Server:  router.domain_not_set.invalid
Address:  192.168.1.254

Non-authoritative answer:
Name:    one.one.one.one
Addresses:  2606:4700:4700::1111
          2606:4700:4700::1001
          1.1.1.1
          1.0.0.1


C:\Users\timni>nslookup dns.google.com
Server:  router.domain_not_set.invalid
Address:  192.168.1.254

Non-authoritative answer:
Name:    dns.google.com
Addresses:  2001:4860:4860::8888
          2001:4860:4860::8844
```

Screenshot website visit via IP address:



Assignment 6.3: subnetting

How many IP addresses are in this network configuration 192.168.110.128/25?

126

What is the usable IP range to hand out to the connected computers?

192.168.110.129 - 254

Check your two previous answers with this Linux command: `ipcalc 192.168.110.128/25`

```
tim@DellG15Tim:~$ ipcalc 192.168.110.128/25
Address:   192.168.110.128      11000000.10101000.01101110.1 0000000
Netmask:   255.255.255.128 = 25 11111111.11111111.11111111.1 0000000
Wildcard:  0.0.0.127            00000000.00000000.00000000.0 1111111
=>
Network:   192.168.110.128/25   11000000.10101000.01101110.1 0000000
HostMin:   192.168.110.129      11000000.10101000.01101110.1 0000001
HostMax:   192.168.110.254      11000000.10101000.01101110.1 1111110
Broadcast: 192.168.110.255      11000000.10101000.01101110.1 1111111
Hosts/Net: 126                   Class C, Private Internet

tim@DellG15Tim:~$
```

Explain the above calculation in your own words.

Hij rekent uit op basis van subnet masker wat alle instellingen zijn en hoeveel hosts er maximaal kunnen. Ook wel ip adres het eerst adres is en welke het laatste. Ook het broadcast adres laat hij zien en wel klasse subnet.

Assignment 6.4: HTML

Screenshot IP address Ubuntu VM:

```
tim@DellG15Tim:~/site$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
fault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group
default qlen 1000
    link/ether 00:15:5d:70:76:d1 brd ff:ff:ff:ff:ff:ff
    inet 172.29.32.22/20 brd 172.29.47.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe70:76d1/64 scope link
       valid_lft forever preferred_lft forever
```

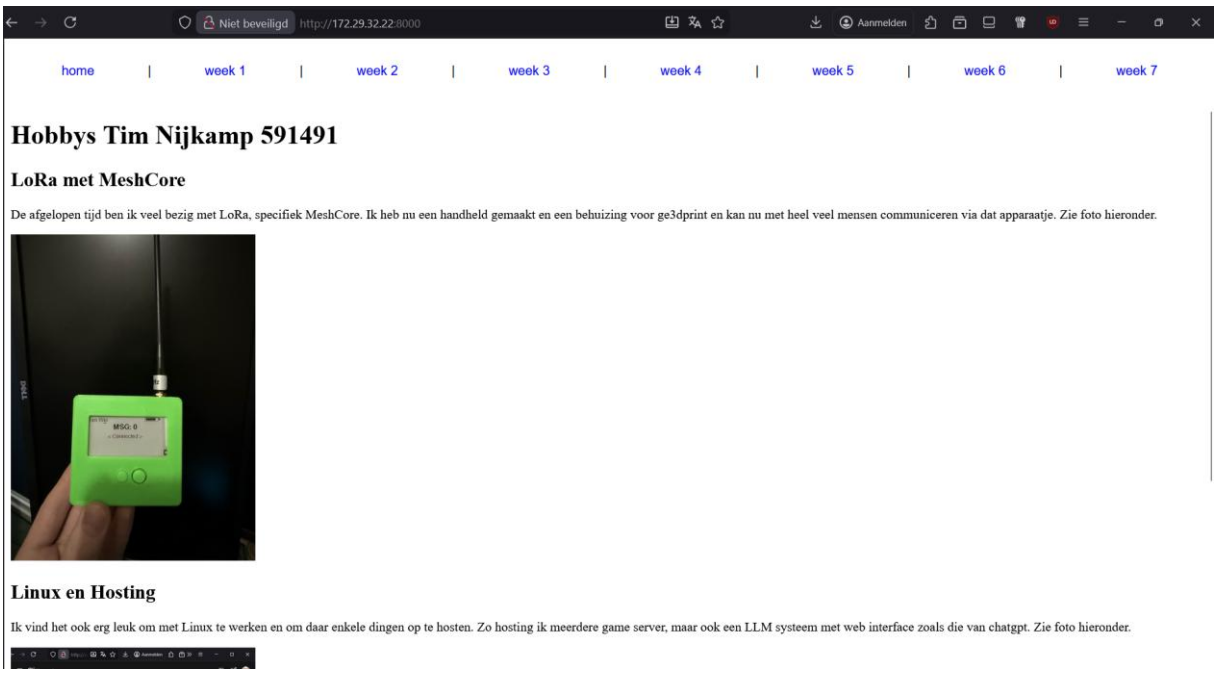
Screenshot of Site directory contents:

```
tim@DellG15Tim:~/site$ ls -la
total 56
drwxrwxrwx 5 tim tim 4096 Jan 13 12:30 .
drwxr-x--- 9 tim tim 4096 Jan 13 12:46 ..
drwxrwxrwx 2 tim tim 4096 Jan 13 12:30 css
-rw-rw-rw- 1 tim tim  903 Jan 13 12:45 home.html
drwxrwxrwx 2 tim tim 4096 Jan 13 12:44 images
-rw-rw-rw- 1 tim tim  637 Jan 13 12:30 index.html
drwxrwxrwx 2 tim tim 4096 Jan 13 12:30 pdf
-rw-rw-rw- 1 tim tim  325 Jan 13 12:30 week1.html
-rw-rw-rw- 1 tim tim  325 Jan 13 12:30 week2.html
-rw-rw-rw- 1 tim tim  325 Jan 13 12:30 week3.html
-rw-rw-rw- 1 tim tim  325 Jan 13 12:30 week4.html
-rw-rw-rw- 1 tim tim  325 Jan 13 12:30 week5.html
-rw-rw-rw- 1 tim tim  325 Jan 13 12:30 week6.html
-rw-rw-rw- 1 tim tim  325 Jan 13 12:30 week7.html
tim@DellG15Tim:~/site$
```

Screenshot python3 webserver command:

```
tim@DellG15Tim:~/site$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
172.29.32.1 - - [13/Jan/2026 12:48:45] "GET / HTTP/1.1" 200 -
172.29.32.1 - - [13/Jan/2026 12:48:45] "GET /css/mypdfstyle.css HTTP/1.1" 20
0 -
172.29.32.1 - - [13/Jan/2026 12:48:45] "GET /home.html HTTP/1.1" 200 -
172.29.32.1 - - [13/Jan/2026 12:48:45] code 404, message File not found
172.29.32.1 - - [13/Jan/2026 12:48:45] "GET /favicon.ico HTTP/1.1" 404 -
172.29.32.1 - - [13/Jan/2026 12:48:45] "GET /images/LoRa.jpg HTTP/1.1" 200 -
172.29.32.1 - - [13/Jan/2026 12:48:45] "GET /images/webui.png HTTP/1.1" 200
-
```

Screenshot web browser visits your site



Assignment 6.5: Network segment

Remember that bitwise java application you've made in week 2? Expand that application so that you can also calculate a network segment as explained in the PowerPoint slides of week 6. Use the bitwise & AND operator. You need to be able to input two Strings. An IP address and a subnet.

IP: 192.168.1.100 and subnet: 255.255.255.224 for /27

```
Example: 192.168.1.100/27
Calculate the network segment
IP Address:   11000000.10101000.00000001.01100100
Subnet Mask:  11111111.11111111.11111111.11100000
```

```
-----------------------------------------------
Network Addr: 11000000.10101000.00000001.01100000

This gives 192.168.1.96 in decimal as the network address.
For a /27 subnet, each segment (or subnet) has 32 IP addresses (2⁵).
The range of this network segment is from 192.168.1.96 to 192.168.1.127.
```

Paste source code here, with a screenshot of a working application.

Ready? Save this file and export it as a pdf file with the name: **week6.pdf**